

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 2

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO2

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 2

Unit Title: Search Techniques, Knowledge
Representation and Reasoning

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Identify the data structure that Depth First Search uses to hold its frontier.

- A) A stack
- B) A queue
- C) A priority queue
- D) A hash table

Q2. State the evaluation function that A* search uses to order its frontier.

- A) $f(n) = g(n) - h(n)$
- B) $f(n) = g(n) + h(n)$
- C) $f(n) = g(n) \times h(n)$
- D) $f(n) = h(n) - g(n)$

Q3. Identify the two parts of a production rule.

- A) A premise and a probability
- B) A slot and a default value
- C) A condition and an action
- D) A node and a labelled link

Q4. State what the knowledge base of an expert system holds.

- A) The facts about the case in hand
- B) The program that applies the rules
- C) The record of the user's questions

D) The rules supplied by the specialists

Q5. Name the element of a semantic network that carries the relation between two concepts.

- A) A slot
- B) A labelled link
- C) A production rule
- D) A clause

Q6. Identify the named compartments in which a frame holds the attributes of a concept.

- A) Links
- B) Clauses
- C) Premises
- D) Slots

Q7. State the name of the logical connective written as an arrow between two propositions.

- A) Implication
- B) Conjunction
- C) Disjunction
- D) Negation

Q8. Name the quantifier of first-order logic that asserts a property holds of every object.

- A) The existential quantifier
- B) The negation operator

- C) The universal quantifier
- D) The implication operator

Q9. Identify the theorem used to revise a probability in the light of new evidence.

- A) Modus ponens
- B) De Morgan's law
- C) Bayes' theorem
- D) The chain rule of logic

Q10. State what the term $P(A)$ represents in Bayes' theorem.

- A) The prior probability of A
- B) The posterior probability of A
- C) The likelihood of the evidence
- D) The probability of the evidence

Q11. Name the term $P(A \text{ given } B)$ in Bayes' theorem.

- A) The prior probability
- B) The likelihood
- C) The marginal probability
- D) The posterior probability

Q12. Identify the search that orders its frontier by the estimated remaining cost alone.

- A) Breadth First Search
- B) Best First Search
- C) Depth First Search
- D) A* search

Q13. State the property of a heuristic that never overestimates the cost remaining to a goal.

- A) Completeness
- B) Optimality
- C) Monotonicity of cost
- D) Admissibility

Q14. Name the local search that moves to the best neighbouring state and halts when no neighbour is better.

- A) Breadth First Search
- B) Best First Search
- C) Hill Climbing
- D) A* search

Q15. State the term for the store of facts an expert system accumulates about the case in hand.

- A) The knowledge base
- B) The working memory
- C) The inference engine

- D) The user interface

Q16. Name the search strategy that expands the shallowest unexpanded node first.

- A) Breadth First Search
- B) Depth First Search
- C) Hill Climbing
- D) Best First Search

Q17. Explain why Depth First Search needs far less memory than Breadth First Search.

- A) It holds no nodes in memory at any time
- B) It holds only the current path and its siblings
- C) It examines fewer nodes in total than BFS
- D) It stores only the nodes at the goal depth

Q18. Distinguish an admissible heuristic from one that is not.

- A) An admissible heuristic never exceeds the true remaining cost
- B) An admissible heuristic never falls below the true cost
- C) An admissible heuristic returns the exact cost always
- D) An admissible heuristic returns the same value everywhere

Q19. Describe what Best First Search ignores that A* takes into account.

- A) The estimated cost remaining to the goal
- B) The number of successors each node has
- C) The depth of the node in the search tree
- D) The cost already incurred in reaching the node

Q20. Interpret the behaviour of A* when its heuristic returns the exact remaining cost at every node.

- A) It expands every node in the whole space
- B) It behaves exactly like Depth First Search
- C) It expands only the nodes on an optimal path
- D) It fails to terminate on any problem at all

Q21. Predict the behaviour of Hill Climbing when it reaches a plateau.

- A) It has no gradient to follow and stalls or wanders
- B) It jumps directly to the global maximum
- C) It reverses and returns to the start state
- D) It reports that no solution can exist

Q22. Recognize the representation described: concepts are drawn as nodes and the relations between them as labelled arrows.

- A) A frame system
- B) A production system
- C) A semantic network
- D) A propositional clause

Q23. Explain what inheritance contributes to a frame system.

- A) A frame copies every fact in working memory
- B) A frame takes its parent's slots unless it overrides them
- C) A frame acquires the rules of the inference engine
- D) A frame receives a probability from its parent

Q24. Summarize what an expert system offers that ordinary software does not.

- A) It runs considerably faster than ordinary software
- B) It requires no knowledge from any human specialist
- C) It improves its own rules from every case it sees
- D) It reasons from stated rules and can explain its conclusion

Q25. Distinguish a constant from a variable in first-order predicate logic.

- A) A variable names one object, a constant ranges over many
- B) Both of the two name exactly one object each
- C) A constant names one object, a variable ranges over many
- D) Both of the two range over many objects

Q26. Explain why A* is described as an informed search.

- A) It uses the labels attached to the training data
- B) It examines every node before choosing one
- C) It requires the solution depth to be known
- D) It uses knowledge of the goal through its heuristic

Q27. Interpret a Bayes calculation in which the prior is very small while the likelihood of the evidence is high.

- A) The posterior can still remain low
- B) The posterior must become close to one
- C) The posterior is unaffected by the prior
- D) The calculation cannot be carried out

Q28. Predict the effect on A* of a heuristic that returns a very large value at every node.

- A) It behaves like Breadth First Search exactly
- B) It behaves like a greedy search and may lose optimality
- C) It refuses to expand any node at all
- D) It is guaranteed to find the cheapest path

Q29. Describe the role of the working memory in a production system.

- A) It holds the rules supplied by the specialists
- B) It holds the program that matches the rules
- C) It holds the questions the user has asked
- D) It holds the facts known about the current case

Q30. Recognize the representation described: a record holding the attributes of one concept in named compartments, with defaults taken from a parent.

- A) A semantic network
- B) A frame
- C) A production rule
- D) A predicate sentence

Q31. Explain why a heuristic function must be inexpensive to compute.

- A) It must be recomputed after every solution
- B) It has to be stored alongside every state
- C) It is evaluated at every node the search considers
- D) It is required to be perfectly accurate

Q32. Distinguish a fact from a rule in an expert system.

- A) A fact states a particular, a rule states a general relation
- B) A rule states a particular, a fact a general relation
- C) Both of the two state particular observations
- D) Both of the two state general relations

Q33. Explain what makes Hill Climbing a local search rather than a systematic one.

- A) It examines only the neighbours of its current state
- B) It examines every state in the whole space
- C) It keeps a complete record of its search tree
- D) It requires an admissible heuristic to run

Q34. Interpret the sentence: for all x, Student(x) implies Enrolled(x).

- A) Some student is enrolled
- B) No student is enrolled
- C) Exactly one student is enrolled
- D) Every student is enrolled

Q35. Predict the result of applying Breadth First Search to a tree with a very large branching factor.

- A) The goal is reached in a constant number of steps
- B) Memory is exhausted long before the goal is reached
- C) The search silently switches to Depth First

D) The search returns a suboptimal solution quickly

Q36. Summarize the applications for which the search techniques of this unit are used.

- A) Training deep neural networks on images
- B) Compressing files for efficient storage
- C) Route planning, puzzle solving and scheduling tasks
- D) Encrypting messages sent over a network

Q37. Describe what the likelihood term contributes to Bayes' theorem.

- A) How probable the hypothesis is before evidence
- B) How probable the evidence is over all hypotheses
- C) How probable the evidence is if the hypothesis holds
- D) How probable the hypothesis is after the evidence

Q38. Explain why the knowledge base of an expert system is kept separate from its inference engine.

- A) The engine then runs considerably faster
- B) The knowledge can be revised without rewriting the program

C) The rules can then be hidden from the user

D) The system then needs no working memory

Q39. Distinguish a semantic network from a frame system by the way each holds the properties of one concept.

- A) A frame spreads them over links, a network gathers them
- B) Both of the two gather the properties into slots
- C) Both of the two spread the properties over links
- D) A network spreads them over links, a frame gathers them in slots

Q40. Explain why probability is used to reason when the available evidence is incomplete.

- A) It expresses degrees of belief that logic cannot
- B) It removes the need for any evidence at all
- C) It guarantees that the conclusion is correct
- D) It converts every rule into a certain fact

----- End of Sample Question Set -----